

Task 4 – Beamforming Simulator for Multi-Domain Applications (Ultrasound, 5G, Radar)

Beamforming and direction of arrival (DOA) are fundamental concepts in modern technologies including wireless communications (e.g., 5G), radar systems, and biomedical imaging such as ultrasound. The core principles rely on phase shifts, time delays, and constructive/destructive interference to steer and shape beams in space.

Description: Develop a comprehensive 2D web-based simulator that demonstrates beamforming concepts across multiple application domains. The simulator should be modular and support at least three independent modes: **Ultrasound Imaging**, **5G Communication**, and **Radar Detection**. The user should be able to explore, visualize, and control beamforming behavior in real time.

General Requirements (Applicable to All Modes)

- The user should be able to:
 - Control the number of transmitters/receivers (probes/elements).
 - Adjust phase shifts and/or time delays for each element in real time.
 - Customize operating frequency/frequencies.
 - Modify the geometry of the array (linear, curved), with full control over curvature parameters.
- Each element (probe) should emit a **visualized signal** (e.g., dotted wave propagation) that reflects realistic spreading behavior.
- The resulting beam (from constructive/destructive interference) should be visualized as a **distinct, emphasized beam profile** (e.g., thicker line or intensity map).
- The simulator should provide **synchronized viewers** showing:
 - Individual element contributions.
 - Combined interference pattern.
 - Final beam direction/profile.
- The user should be able to steer the beam dynamically by modifying delays/shifts.
- Support multiple phased array units operating simultaneously.
- Provide parameter setting files for predefined scenarios that can be loaded, modified, and saved.

Mode 1: Ultrasound Beamforming Simulator

- Simulate an ultrasound probe consisting of multiple transducer elements.
- The user should be able to:
 - Customize probe geometry (linear or curved array).
 - Control number and spacing of elements.

- Adjust firing timing (delays) and phase shifts for each element.
- Each probe element should emit a **wavefront visualization** representing ultrasound propagation.
- The system should:
 - Show the formation of the final beam as the summation of all emitted signals.
 - Allow real-time steering and focusing of the beam by modifying firing delays.
- The visualization should clearly distinguish between:
 - Individual element emissions.
 - The resulting focused beam.
- Think of how to represent realistic propagation and attenuation behavior.

Mode 2: 5G Beamforming and User Tracking Simulator

- Develop an interactive 2D environment (game-like interface) where:
 - The user can place multiple 5G base stations (towers).
 - The user can add/remove multiple mobile users in the environment.
- Each tower should:
 - Contain a customizable number of antenna elements.
 - Have a defined coverage range (visualized).
- Users should be able to:
 - Select a mobile user and control its movement dynamically (e.g., via keyboard).
- The system should:
 - Form and steer beams toward active users using beamforming principles.
 - Visually show:
 - * Individual antenna emissions.
 - * Final directed beam toward each user.
- Multi-user handling:
 - If multiple users are within range:
 - * The antenna elements should be **distributed dynamically** among users.
 - * Each subset of elements forms an independent beam.
- Connectivity logic:
 - A user connects to only one tower at a time.
 - If overlapping coverage exists:
 - * The user remains connected to the current tower until disconnection conditions are met.
 - * Then connection may switch to another tower.

- Think carefully about how to simulate:
 - Resource allocation (elements per user),
 - Connection management,
 - Real-time beam steering.

Mode 3: Radar Beamforming Simulator

- Simulate a radar system based on phased array beam steering.
- The system should:
 - Display a circular detection area.
 - Continuously scan the environment using a **steered beam** (not a simple rotating line).
- The beam movement should be achieved by:
 - Dynamically adjusting delays/phases of array elements (beamforming-based scanning).
- The user should be able to:
 - Control scan speed and direction.
 - Adjust array parameters and observe their effect on detection.
- Objects/targets should be placed in the environment, and the system should:
 - Detect targets when they fall within the beam.
 - Visually indicate detection events.
- Compare traditional “rotating line radar visualization” with beamforming-based steering.

Advanced Requirements

- Do your research to:
 - Compare classical beamforming approaches with any AI-based beam steering or optimization methods (if applicable).
 - Discuss limitations and trade-offs between different configurations (array geometry, frequency, number of elements).
- Provide at least three predefined scenarios across the different modes (e.g., medical imaging, urban 5G deployment, radar tracking).

Code Practice

1. Avoid code repetition. Use modular design and reusable components across the three modes.
2. Apply Object-Oriented Programming (OOP) principles:
 - Encapsulate array elements, signals, and visualization logic.
 - Separate simulation logic from UI handling.
3. The UI should be intuitive and responsive:

- All changes should reflect immediately without requiring manual refresh.
4. You will be penalized for:
- Bugs or crashes,
 - Poor performance in real-time interaction,
 - Unhandled edge cases,
 - Unused or redundant code.